MLP Project Milestone 4: Model Building and Hyperparameter Exploration

To understand about all the milestones in detail, Please *click here*

Roll number	

Submission guidelines

1. **Make your own copy with File > Make a copy, and write only in your copy.**
2. **Once you have finished writing your report, save or export the .docx document as a PDF.**
3. **Rename the PDF file using the following exact convention: [roll-number]_26t3_M[Milestone-Number].pdf**
4. **Example: If your roll number is 21f1001234 and you are submitting the PDF for Milestone 4, your file must be named exactly: 21f1001234_26t3_M4.pdf**

Instructions

- Use the Project data to answer the questions.
- Write each answer in the "Your answer" box under the question, and paste the Python code (wherever required) you ran into the "Your code" box as text, not as a screenshot.
- Support every claim with evidence from your own analysis of the competition data: a number, a small table or a plot (Insert > Image). Summarise long outputs instead of pasting them in full.
- The points under each question describe what an answer should cover. If a point does not apply to your dataset (for example, there are no date columns), you can skip it.
- Write in your own words. Clear reasoning backed by evidence matters more than length of your answer.
- There is no single correct answer, these questions are subjective and there can be multiple valid explanations of questions. Use your interpretation and understanding to answer the questions.
- You need not include all the columns(features) while answering questions, you can select a few meaningful columns and do analysis on them. Try to cover at least 50% of the columns(features) for the questions in this milestone.
- You MAY use a different kaggle notebook instead of the one shared with the host to do the task given in milestones.
- The answer length can vary depending on questions. Keep your answer concise and give a summary of the analysis and cover details using visualizations/bullet points wherever required. The maximum length of answer can be considered around 100 to 400 words.
- **Plots, graphs and figures:** add them in the "Figures" section at the end of this document, under the heading of the question they belong to, not inside the answer boxes. Number each figure as Figure Q<question number>.<figure number> (for example, Figure Q3.1) and refer to it by this number in your answer (for example, "see Figure Q3.1").

Q1. Model Building Plan

Identify suitable models for the given prediction problem and explain the planned model-building approach.

One suggested approach would be as follows:

- Identify suitable linear, non-tree, tree-based, and ensemble models for the prediction problem, where applicable.
- Justify the selection of models based on the nature of the target variable, dataset characteristics, and relevant modelling assumptions.

Write your answer here.

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# Paste the code you ran for this question
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Q2. Linear Models

Explore linear models and describe how their assumptions and characteristics influence their behavior on the given dataset. Implement and explore at least one suitable model, identify and describe the role of hyperparameters of the model.

One suggested approach would be as follows:

- Explain the fundamental assumptions of the selected linear model and how it represents the relationship between input features and the target variable.
- Discuss the impact of preprocessing steps on the model. Describe the strength and limitations of the model based on the dataset.
- Explain the role of relevant hyperparameters of the selected model and describe how they influence the model.

Write your answer here.

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Q3. Other Non-Tree Models

Build a suitable non-tree and non linear model for the given prediction problem. Choose at least one model and describe the underlying assumptions, mechanisms and behavior on the given dataset. Discuss the impact of preprocessing, identify and describe the role of hyperparameters of the model.

One suggested approach would be as follows:

- Explain the fundamental assumptions of the selected model and how it represents the relationship between input features and the target variable.
- Discuss the impact of preprocessing steps on the model. Describe the strength and limitations of the model based on the dataset.
- Explain the role of relevant hyperparameters of the selected model and describe how they influence the model.

Write your answer here.

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Q4. Decision Trees

Explore decision trees and understand how they learn from data. Describe the underlying assumptions, mechanisms and behavior on the given dataset. Discuss the impact of preprocessing, identify and describe the role of hyperparameters of the model.

One suggested approach would be as follows:

- Explain the process of the decision tree model and how it represents the relationship between input features and the target variable.
- Discuss the impact of preprocessing steps on the model. Describe the strength and limitations of the model based on the dataset.
- Explain the role of relevant hyperparameters of the selected model and describe how they influence the model.

Write your answer here.

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Q5. Bagging Ensembles

Explore bagging-based ensemble models and understand how they learn from data. Describe the underlying mechanisms and behavior of these models on the given dataset. Discuss the impact of preprocessing, identify and describe the role of hyperparameters of the selected model.

One suggested approach would be as follows:

- Explain the working mechanism of the selected bagging model and how it combines multiple base models to make predictions.
- Discuss the impact of preprocessing steps on the model. Describe the strength and limitations of the model based on the dataset.
- Explain the role of relevant hyperparameters of the selected model and describe how they influence the model.

Write your answer here.

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Q6. Boosting Ensembles

Explore boosting-based ensemble models and understand how they learn from data. Describe the underlying mechanisms and behavior of these models on the given dataset. Discuss the impact of preprocessing, identify and describe the role of hyperparameters of the selected model.

One suggested approach would be as follows:

- Explain the working mechanism of the selected boosting model and how it is built.
- Discuss the impact of preprocessing steps on the model. Describe the strength and limitations of the model based on the dataset.
- Explain the role of relevant hyperparameters of the selected model and describe how they influence the model.

Write your answer here.

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Q7. Voting and Stacking

Explore voting and stacking ensemble models and understand how the algorithm works. Describe the underlying mechanisms and behavior of these models on the given dataset. Discuss the impact of preprocessing, identify and describe the role of hyperparameters of the selected model.

One suggested approach would be as follows:

- Explain the working mechanisms of voting and stacking models and how they combine predictions from multiple base models to make final predictions.
- Discuss the impact of preprocessing steps on the model. Describe the strength and limitations of the model based on the dataset.
- Explain the role of relevant hyperparameters of the selected model and describe how they influence the model.

Write your answer here.

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Q8. Hyperparameter Tuning(HPT) Strategy

Explain the process of selecting suitable hyperparameter configurations for machine learning models. Explore how cross-validation is used to obtain reliable estimates of model performance and how different search strategies can be used to identify suitable configurations.

One suggested approach would be as follows:

- Explain the factors that should be considered when selecting hyperparameters and defining their search ranges. Discuss how the choice of hyperparameter values can influence the trade-off between model complexity and generalisation.
- Explore different cross-validation strategies and describe the importance and implementation steps of them.
- Explore HPT approaches. Explain how they explore the search space and compare their advantages, limitations, and computational requirements.

Write your answer here.

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Q9. Hyperparameter tuning Effects

Perform hyperparameter tuning on at least three machine learning models to identify suitable hyperparameter configurations for the given prediction problem. Apply appropriate cross-validation and search strategies to evaluate different configurations.

One suggested approach would be as follows:

- Identify the important hyperparameters of models and define suitable ranges or candidate values for tuning. Explain the reasoning behind the selected search spaces.
- Choose an appropriate hyperparameter search strategy. Justify the choice based on the search space and computational requirements.
- Select an appropriate cross-validation strategy for each model and use it to evaluate different hyperparameter configurations.
- Report the best hyperparameter configuration identified for each model along with its corresponding cross-validation score. Describe the observations from the tuning process.

Write your answer here.

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Q10. Model Score Analysis

Compare the scores obtained from the initial models and the models after hyperparameter tuning.

One suggested approach would be as follows:

- Present the scores before and after hyperparameter tuning for the selected models.
- Identify which models showed improvement after tuning and describe the observed changes.

Write your answer here.

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# Paste the code you ran for this question
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Figures

Add all plots, graphs and figures for this milestone in this section, under the heading of the question they belong to.

How to add a figure

1. Place your cursor below the heading of the question the figure belongs to, and use Insert > Image to add the figure.
2. Number each figure in the format Figure Q<question number>.<figure number>. For example, the first two figures for Q3 are Figure Q3.1 and Figure Q3.2.
3. Write the figure number and a short title on the line just above the figure, for example: Figure Q9.1: Validation score for different values of a hyperparameter.
4. In your answer, refer to the figure by its number, for example: "The score stops improving beyond a certain value of the hyperparameter (see Figure Q9.1)."
5. Resize each figure so that it fits within the page width and all axis labels and legends are readable.
6. Replace the placeholder lines under each heading with your figures. If a question has no figures, leave its heading as it is.

Figures for Q1. Model Building Plan

Figure Q1.1: [Short title of the figure]

[Insert image here]

Figures for Q2. Linear Models

Figure Q2.1: [Short title of the figure]

[Insert image here]

Figures for Q3. Other Non-Tree Models

Figure Q3.1: [Short title of the figure]

[Insert image here]

Figures for Q4. Decision Trees

Figure Q4.1: [Short title of the figure]

[Insert image here]

Figures for Q5. Bagging Ensembles

Figure Q5.1: [Short title of the figure]

[Insert image here]

Figures for Q6. Boosting Ensembles

Figure Q6.1: [Short title of the figure]

[Insert image here]

Figures for Q7. Voting and Stacking

Figure Q7.1: [Short title of the figure]

[Insert image here]

Figures for Q8. Hyperparameter Tuning Strategy

Figure Q8.1: [Short title of the figure]

[Insert image here]

Figures for Q9. Hyperparameter Effects and Configuration Rationale

Figure Q9.1: [Short title of the figure]

[Insert image here]

Figures for Q10. Model Score Analysis

Figure Q10.1: [Short title of the figure]

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